

Real-Time Network Quality & Customer Experience Analytics

Introduction

This document describes a real-time analytics solution for monitoring telecom network quality and customer experience.

The system ingests streaming Call Detail Records (CDRs), enriches them with customer data, computes key performance indicators (KPIs), and presents insights through interactive dashboards.

It is designed to demonstrate how organisations can detect network issues early, understand customer impact, and enable data-driven operational decisions in near real time.

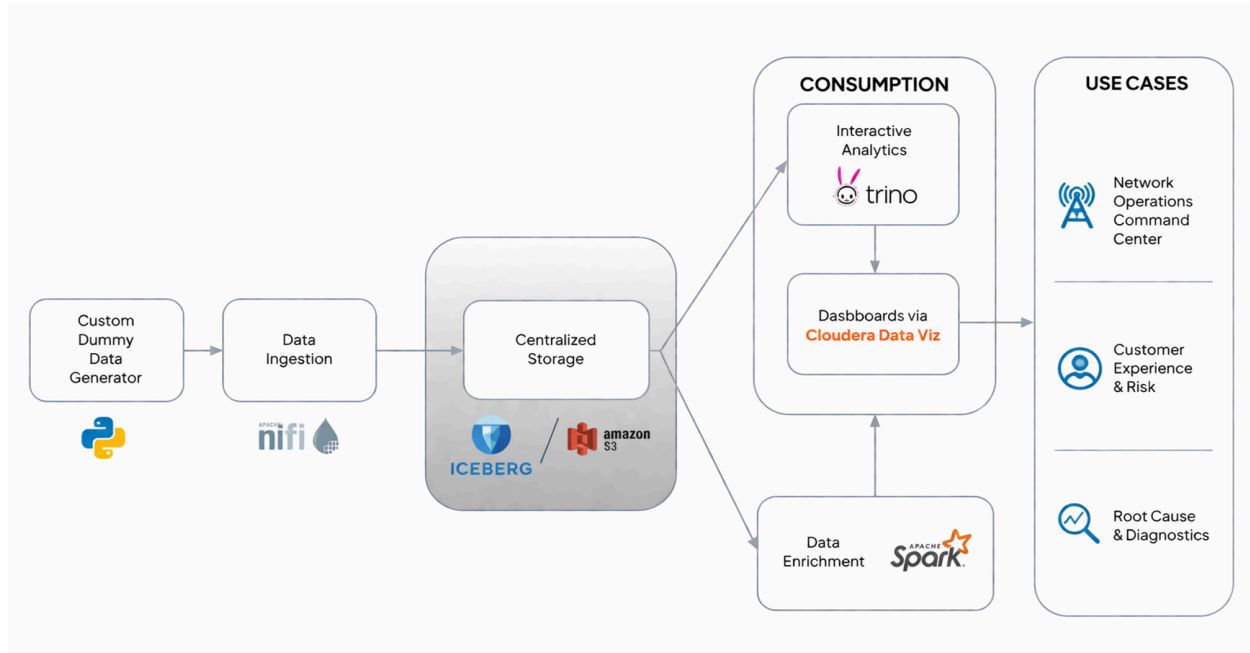
Objective

Design and implement an end-to-end data pipeline that:

- Ingests **real-time Call Detail Records (CDRs)**
- Enriches them with **customer profile data**
- Computes **network KPIs**
- Enables **near real-time analytics and dashboards**

Architecture Overview

The solution processes telecom data in real time using a streaming architecture, enabling continuous ingestion, transformation, and visualisation of network performance metrics.



Key Components

1. Data Generation Layer

- A Python-based generator (Flask API) simulates telecom CDR data
- Each record includes: `event_id`, `timestamp`, `caller_id`, `receiver_id`, `duration`, `cell_tower_id`, `call_result` (SUCCESS/DROPPED), `signal_strength`

2. Ingestion Layer

Tool: Apache NiFi

- NiFi invokes the API using `InvokeHTTP`
- Data is written to S3 in Parquet format

3. Processing Layer (Spark on CDE)

There're three Spark jobs created for different purposes.

a. CDR Ingestion

- NiFi pulls data from application API
- Writes Parquet files to S3 landing zone
- Finally using Sql add data to external table: **telco_fact.cdr_raw**

b. CDR Enrichment

- Joins raw CDR data with customer_profile table created using dataset (Kaggle)
- Adds derived business attributes
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Attribute	Description
customer_tier	Gold / Silver / Bronze
experience_score	Based on signal strength & call success
network_quality	GOOD / AVERAGE / POOR
high_risk_customer	Based on churn + network issues

- Output Table: **telco_fact.cdr_enriched**

NOTE: Derived metrics are computed incrementally using streaming, avoiding full data scans.

c. KPI Computation

- Performs **1-minute window aggregation**
- Computes key metrics:

Metric	Description
drop_rate_pct	% of dropped calls
avg_signal_strength	Average signal strength
avg_experience_score	Customer experience score
high_risk_customers	Count of impacted users

- Output table: **telco_fact.kpi_network_quality**

NOTE: Derived metrics are computed incrementally using streaming, avoiding full data scans.

4. Storage Layer

Format: Apache Iceberg

- All datasets are stored as Iceberg tables
- Benefits:
 - Schema evolution
 - Time-travel support
 - Efficient large-scale analytics
 - Optimised for streaming + batch workloads

5. Query Layer

Tool: Trino

- Provides a fast SQL interface over Iceberg tables
- Enables direct integration with BI tools

6. Visualisation Layer

Tool: Cloudera Data Visualization (CDV)

Dashboards created:

- Network Operations Command Center
- Customer Experience & Risk
- Root Cause & Diagnostics

End-to-End Data Flow

```
HTML
CDR Generator (Flask API)
  ↓
Apache NiFi (InvokeHTTP)
  ↓
S3 (Parquet data landing)
  ↓
Spark Streaming (Enrichment) ← customer_profile (static)
  ↓
Iceberg: cdr_enriched
  ↓
Spark Streaming (KPI Aggregation)
  ↓
Iceberg: kpi_network_quality
  ↓
Trino (SQL Layer)
  ↓
CDV Dashboards

[Supporting Layer]
S3 → SQL External Table → cdr_raw
```